

ME 333 - MECHANICS OF MATERIALS PRACTICE PROBLEMS

(1) Given the following stress functions:

$$\phi_1 = C_1x^2 + C_2xy + C_3y^2$$

$$\phi_2 = C_1x^3 + C_2x^2y + C_3xy^2 + C_4y^3$$

- (a) Check if these stress functions satisfy plane problems.
- (b) Obtain stress fields for ϕ_1 and ϕ_2 .

Ans: (a) both Airy stress functions satisfy biharmonic equation and hence the plane problems,
 (b) ($\sigma_{xx} = 2C_3$; $\sigma_{yy} = 2C_1$; $\sigma_{xy} = -C_2$); ($\sigma_{xx} = 2C_3x + 6C_4y$; $\sigma_{yy} = 6C_1x + 2C_2y$; $\sigma_{xy} = -2C_2x - 2C_3y$)

(2) Given:

$$\sigma_{xx} = Px^3y - 2C_1xy + C_2y$$

$$\sigma_{yy} = Pxy^3 - 2Px^3y$$

$$\sigma_{xy} = -\frac{3}{2}Px^2y^2 + C_1y^2 + \frac{1}{2}Px^4 + C_3$$

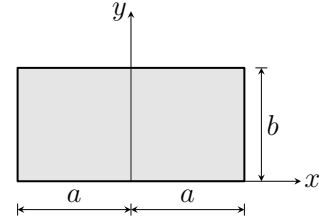


FIGURE 1

- (a) Verify this is a solution for a thin plate of thickness t ,
- (b) obtain the corresponding stress function,
- (c) find resultant normal and shear boundary forces along the edges $y = 0$ and $y = b$ of the plate.

Ans: (a) Thin plate corresponds to a plane stress condition - equilibrium and Laplacian of sum of in-plane normal stresses being zero have to be checked,

(b) The Airy stress function can be obtained by solving the biharmonic equation and results in:

$$\Phi = \frac{Px^3y^3}{6} - \frac{Px^5y}{10} + (C_4y^3 + C_5y^2 + C_6y + C_7)x + (C_8y^3 + C_9y^2 + C_{10}y + C_{11}),$$

(c) $V_x|_{(y=0)} = \frac{Pa^5t}{5} + 2C_3at$, $P_y|_{(y=0)} = 0$, $V_x|_{(y=b)} = -Pa^3t \left(b^2 - \frac{a^2}{5} \right) + 2at(C_1b^2 + C_3)$, $P_y|_{(y=b)} = 0$. Note that P_y and V_x are the normal and shear forces, respectively.

(3) Derive the corresponding displacement fields for thick-walled cylinders in case:

- (a) $p_o = 0$ (pressure on the outer periphery @ $r = R_o$)
- (b) $p_i = 0$ (pressure on the inner periphery @ $r = R_i$)
- (c) both p_i and p_o exist

Ans: The displacement field is:

$$u = \frac{(1-\nu)r}{E} \left(\frac{P_i R_i^2 - P_o R_o^2}{R_o^2 - R_i^2} \right) + \frac{(1+\nu)}{rE} \left(\frac{R_i^2 R_o^2 (P_i - P_o)}{R_o^2 - R_i^2} \right)$$

The above field expression is for (c) and results for (a) and (b) can be obtained by considering only external pressure or only internal pressure, respectively.

- (4) Given a composite cylinder where a rigid solid cylinder is bounded by a cylinder of elastic material and is subjected to an external pressure p_o , as shown in Fig.2, derive the stress and strain fields for the outer cylinder.

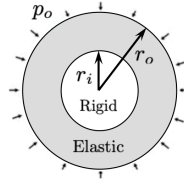


FIGURE 2. A composite rigid inner and elastic thick walled outer cylinder

Ans: Assuming perfect interface and $u = 0 @ r = r_i$ considering the rigid inner cylinder, $p_i = p_o \left(\frac{2\nu r_o^2}{r_o^2 - (1 - 2\nu)r_i^2} \right)$ can be derived to be the internal pressure experienced by the outer cylinder. Once p_i is known then the displacement, strain and stress fields can be obtained using the equations derived and discussed during the lectures.

- (5) A thin walled circular cylindrical vessel of diameter d and wall thickness t is subjected to an internal pressure p_i . Given a small circular hole in the vessel wall show that the maximum tangential and axial stresses at the hole are $\sigma_t = \frac{5pd}{4t}$, $\sigma_a = \frac{pd}{4t}$, respectively.